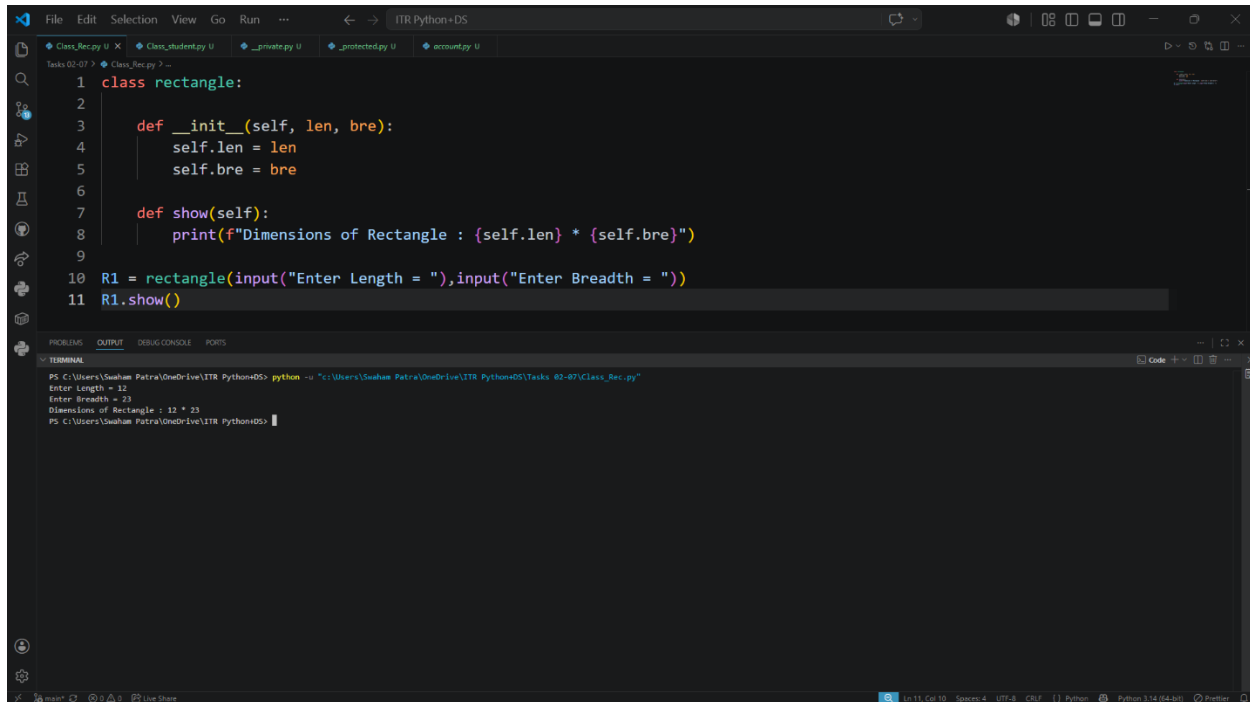


Swaham Pitambar Patra

Task 1:

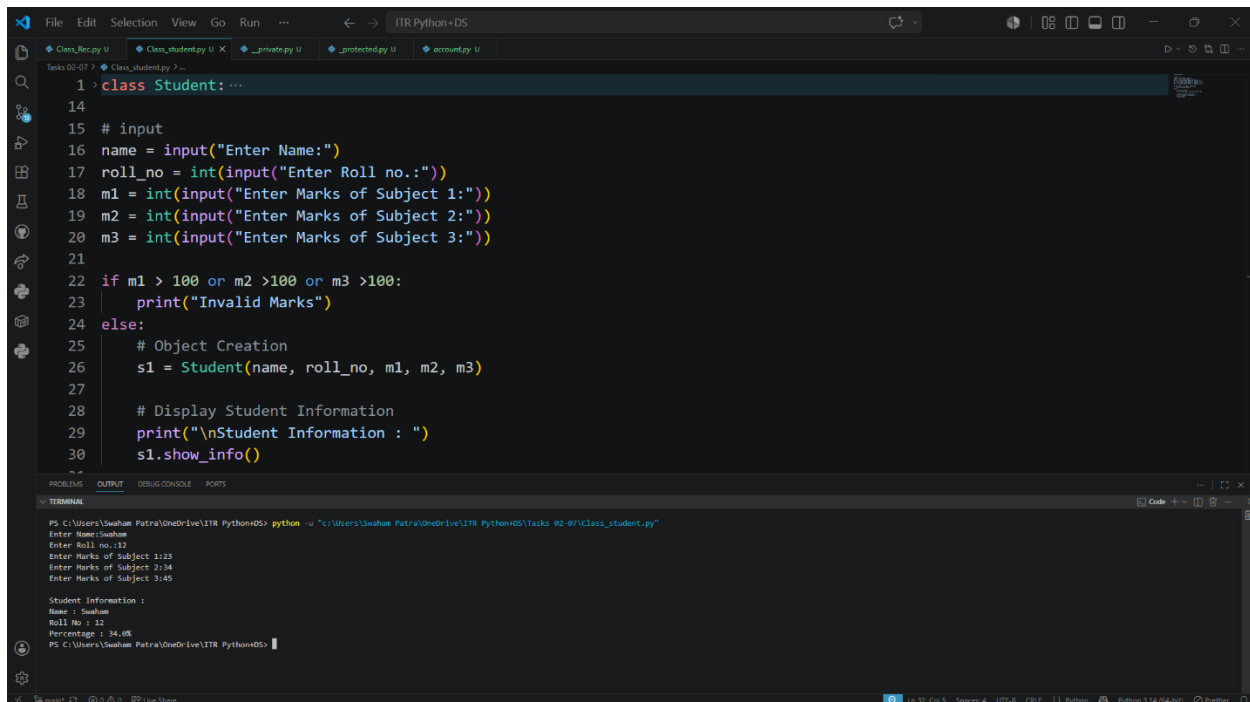


```
1 class rectangle:
2
3     def __init__(self, len, bre):
4         self.len = len
5         self.bre = bre
6
7     def show(self):
8         print(f"Dimensions of Rectangle : {self.len} * {self.bre}")
9
10 R1 = rectangle(input("Enter Length = "),input("Enter Breadth = "))
11 R1.show()
```

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR PythonDS> python -u "C:\Users\Swaham Patra\OneDrive\ITR PythonDS\Tasks 02-07\Class_Rec.py"
Enter Length = 12
Enter Breadth = 23
Dimensions of Rectangle : 12 * 23
PS C:\Users\Swaham Patra\OneDrive\ITR PythonDS>
```

Task 2:



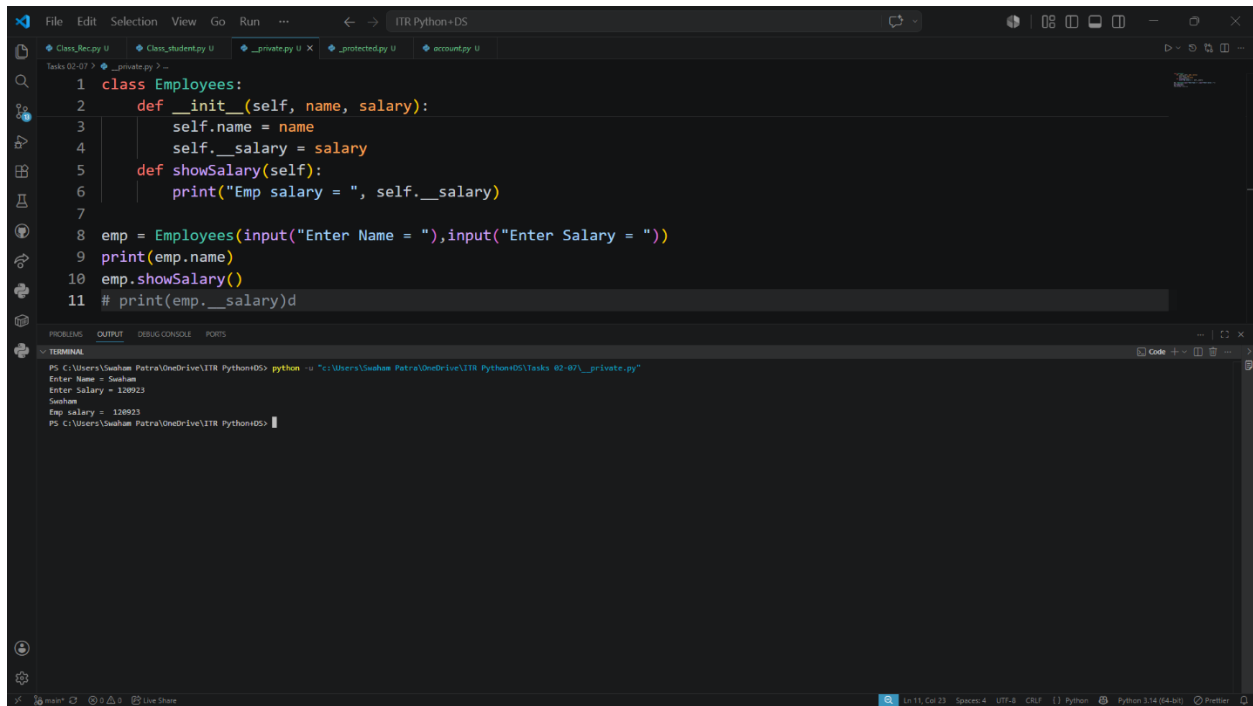
```
1 > class Student: ...
14
15 # input
16 name = input("Enter Name:")
17 roll_no = int(input("Enter Roll no.:"))
18 m1 = int(input("Enter Marks of Subject 1:"))
19 m2 = int(input("Enter Marks of Subject 2:"))
20 m3 = int(input("Enter Marks of Subject 3:"))
21
22 if m1 > 100 or m2 > 100 or m3 > 100:
23     print("Invalid Marks")
24 else:
25     # Object Creation
26     s1 = Student(name, roll_no, m1, m2, m3)
27
28     # Display Student Information
29     print("\nStudent Information : ")
30     s1.show_info()
```

TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR PythonDS> python -u "C:\Users\Swaham Patra\OneDrive\ITR PythonDS\Tasks 02-07\Class_student.py"
Enter Name:Swaham
Enter Roll no.:12
Enter Marks of Subject 1:23
Enter Marks of Subject 2:34
Enter Marks of Subject 3:45

Student Information :
Name : Swaham
Roll No : 12
Percentage : 34.4%
PS C:\Users\Swaham Patra\OneDrive\ITR PythonDS>
```

Task 3:



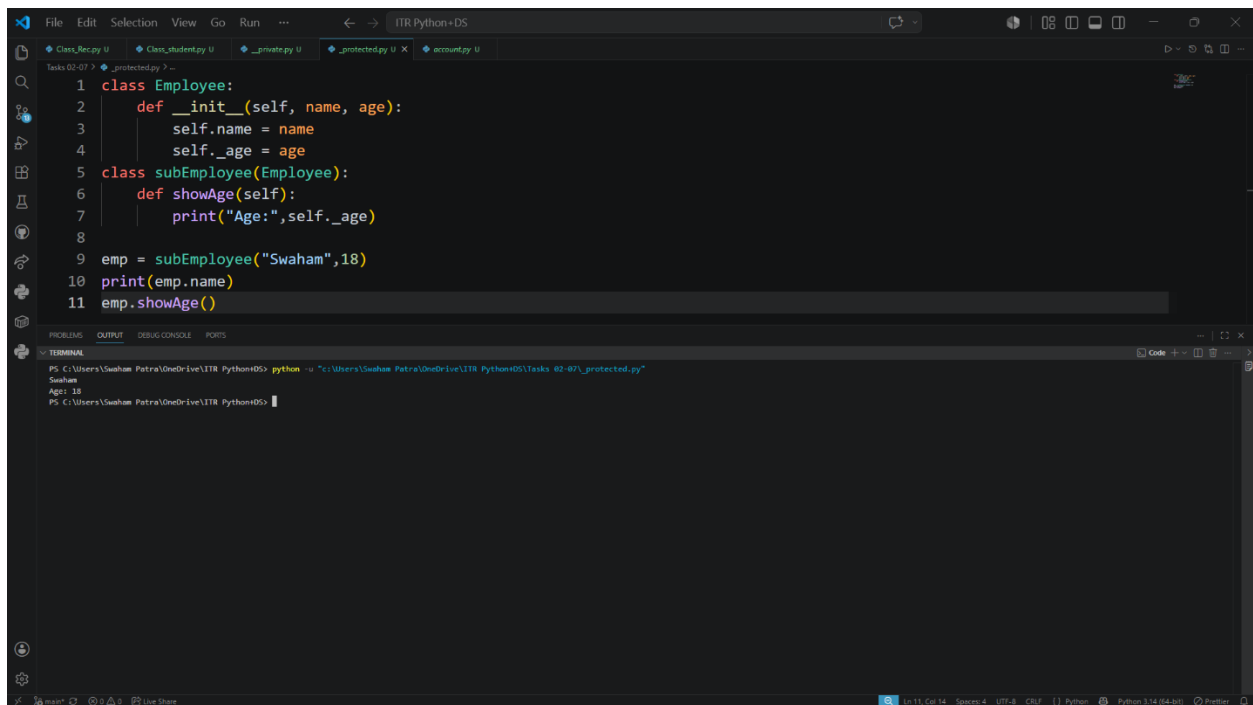
The screenshot shows a VS Code editor window with a Python file named `_private.py`. The code defines a class `Employees` with an `__init__` method and a `showSalary` method. It then creates an instance `emp` and prints its name and salary.

```
1 class Employees:
2     def __init__(self, name, salary):
3         self.name = name
4         self.__salary = salary
5     def showSalary(self):
6         print("Emp salary = ", self.__salary)
7
8 emp = Employees(input("Enter Name = "),input("Enter Salary = "))
9 print(emp.name)
10 emp.showSalary()
11 # print(emp.__salary)d
```

The terminal output shows the execution of the script, where the user enters 'Swaham' for the name and '120923' for the salary. The output is:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python\DS\tasks 02-07\_private.py"
Enter Name = Swaham
Enter Salary = 120923
Swaham
Emp salary = 120923
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS>
```

Task 4:



The screenshot shows a VS Code editor window with a Python file named `_protected.py`. The code defines a base class `Employee` with an `__init__` method and a `showAge` method. It then creates a subclass `subEmployee` that inherits from `Employee` and overrides the `showAge` method. It then creates an instance `emp` of `subEmployee` and prints its name and age.

```
1 class Employee:
2     def __init__(self, name, age):
3         self.name = name
4         self._age = age
5 class subEmployee(Employee):
6     def showAge(self):
7         print("Age:",self._age)
8
9 emp = subEmployee("Swaham",18)
10 print(emp.name)
11 emp.showAge()
```

The terminal output shows the execution of the script, where the user enters 'Swaham' for the name and '18' for the age. The output is:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python\DS\tasks 02-07\_protected.py"
Swaham
Age: 18
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS>
```

Task 5:

```
File Edit Selection View Go Run ... ITR Python+DS
1 class BankAccount:
2
3     def __init__(self, account_holder, accountNo, balance):
4         self.account_holder = account_holder
5         self.accountNo = accountNo
6         self.__balance = balance
7
8     def deposit(self, amount):
9         self.__balance += amount
10        print(f"Deposited (amount). New balance: {self.__balance}")
11
12    def Withdraw(self, amount):
13        if amount <= self.__balance:
14            self.__balance -= amount
15            print(f"Withdraw (amount). New balance: {self.__balance}")
16        else:
17            print("Insufficient funds.")
18
19    def DisAcc_Details(self):
20        print(f"Account Holder: {self.account_holder}")
21        print(f"Account Number: {self.accountNo}")
22        print(f"Balance: {self.__balance}")
23
24    print("Access Account:")
25    name = input("Enter Name:")
26    AccNo = int(input("Enter Account Number:"))
27    CurrentBalance = float(input("Enter Deposited Amount:"))
28
29    C1 = BankAccount(name, AccNo, CurrentBalance)
30
31    option = int(input("Enter 1 for Deposit, 2 for Withdraw, 3 for Display Account Details: "))
32
33    if option == 1:
34        C1.deposit(float(input("Enter Amount to Deposit: ")))
35    elif option == 2:
36        C1.Withdraw(float(input("Enter Amount to Withdraw: ")))
37    elif option == 3:
38        C1.DisAcc_Details()
39    else:
40        print("Invalid option!")
41
```

Outputs:

```
File Edit Selection View Go Run Terminal ... ITR Python+DS
PROBLEMS OUTPUT DEBUG CONSOLE PORTS
TERMINAL
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 02-07\_protected.py"
Swaham
Age: 18
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python+DS\Tasks 02-07\problemSTMT.py"
Access Account:
Enter Name:Swaham
Enter Account Number:1234456
Enter Deposited Amount:12
Enter 1 for Deposit, 2 for Withdraw, 3 for Display Account Details: 3
Account Holder: Swaham
Account Number: 1234456
Balance: 12.0
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
Ln 35, Col 18 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Prettier
```

